



Dany Katamba Mpoyi

📞 Phone number: (+39) 3802396750 ✉ Email address: katambadany@gmail.com

📍 Home: via de Viti de Marco,24, 70125 Bari (Italy)

ABOUT ME

Research engineer specializing in NDT, SHM,PHM, CbM for structures, materials, and rotating machinery, combining signal processing, machine learning, and deep learning for intelligent diagnostics and prognostics. Experienced in multi-sensor data, vibration and acoustic emission monitoring, and experimental system integration.

WORK EXPERIENCE

🏢 **Politecnico di Bari** – Bari, Italy

City: Bari | Country: Italy

Links https://www.poliba.it/sites/default/files/aassegni/dr21_approvazione_atti-signed.pdf | https://www.poliba.it/sites/default/files/aassegni/dr21_approvazione_atti-signed.pdf | https://www.poliba.it/sites/default/files/aassegni/dr21_approvazione_atti-signed.pdf

Postdoctoral Researcher (I-MOBI Project, Grant n. CUP B99J24001010005)

[07/01/2025 – Current]

- Contributed to R&D on design and testing methodologies for materials and vehicle components, emphasizing energy efficiency, CO2 reduction, and lightweight structure.
- Performed mechanical characterization (static, fatigue, and impact tests) on chassis materials, collecting high-frequency experimental data and conducting numerical analysis (linear static and dynamic FEM simulations) to model the frame behavior under operational stresses.
- Correlated **experimental and numerical results** to validate material models and identify **critical structural areas** for potential **lightweight remodeling**.
- Developed AI-based predictive frameworks using machine learning to model material behavior, support data-driven decisions, and optimize resistance/weight ratios in vehicle parts .
- Contributed to the preparation of a **scientific report**.
- **Collaborated with industrial partners and research centers** involved in the I-MOBI project to align experimental findings and simulation outcomes with real design and manufacturing requirements.

PhD Researcher (Project IN MOOSHEAC, Grant n. CUP D53D23003600006)

[01/02/2022 – 01/10/2024]

- Engaged in advanced research on self-healing composites funded under the National Recovery and Resilience Plan (PNRR).
- Conducted experimental testing of self-healing composite materials with deep learning-based AE waveform analysis for real-time damage detection and classification.
- Integrated data mining and AI-driven frameworks for prognostics, enabling early detection of damage evolution and health indicator extraction in aerospace applications.
- Validated predictive systems to enhance structural reliability and extend service life, with focus on multi-sensor integration (AE, vibration)
- Published scientific findings on advanced composite materials, acoustic emission techniques, and the application of AI in predictive damage monitoring.

Researcher, RIPARTI Project (Grant No. CUP-D93C23000100001)

[01/01/2023 – 01/09/2025]

- Acted as the **main contributor** in advanced studies on intelligent monitoring systems for industrial customized vehicles.
- Led the development of vibration-based fault diagnosis methods in rotating joints using signal processing and machine learning techniques.
- Designed and implemented deep learning models (CNN architectures) integrated with kurtogram-based spectral analysis.
- Investigated sensor placement optimization and multi-sensor data fusion strategies for real-time predictive maintenance.
- Collaborated with academic and industrial partners and contributed to scientific documentation and project dissemination.

 **Politecnico di Bari** – Bari, Italy

City: Bari | Country: Italy

Link <https://doi.org/10.5194/egusphere-egu25-13496>

PhD Researcher (Project TUBES, Grant n. CUP D53D2302292000)

[01/11/2023 – 31/10/2024]

- Applied advanced acoustic emission (AE) methods for hazard assessment in lava tube structures, involving signal processing and multi-sensor data fusion for structural evaluation.
- Designed AI-based predictive models for behavior modeling and risk mitigation, integrating experimental data acquisition under dynamic conditions.
- **Contributed to the preparation of a scientific report** on *AE-based monitoring and predictive of volcanic subsurface structures*.
- **Collaborated with multiple universities and the Istituto Nazionale di Geofisica e Vulcanologia (INGV)** to coordinate data acquisition campaigns, algorithm validation, and cross-disciplinary interpretation of AE and geophysical datasets.

 **Bralima Heineken Group** – Kinshasa , Democratic Republic of the Congo

City: Kinshasa | Country: Democratic Republic of the Congo

Packaging engineer

[09/11/2019 – 01/11/2021]

- Lead major incident resolution, coordinating response efforts across technical teams to ensure business service are restored rapidly.
- Manage ticketing systems (JIRA) to document incidents, track timelines, and assign priority levels based on business impact.
- Troubleshoot facility and rack-level events and resolve within established SLAs.
- Collaborate with senior engineers n moderate complexity projects aimed at reducing operational costs while improving facility performance.
- Perform operational readings, rack installs, and decommissioning of racks while ensuring adherence to safety protocols.
- Managed vendors and contractors on-site, ensuring all work met established procedures and local legislation.
- Provide on-call support to resolve critical incidents during non-business hours.
- Lead APM teams and continuously train operators on safety, adherence to standards, SOP compliance, and accurate operational reporting.

 **Institut Supérieur des Techniques Appliquées (ISTA/Kinshasa)** – Kinshasa , Democratic Republic of the Congo

City: Kinshasa | Country: Democratic Republic of the Congo

Lecturer in Industrial Maintenance

[01/10/2019 – 04/08/2021]

- Applied fundamental and advanced principles of **industrial maintenance** within large-scale production systems.
- **Taught and implemented maintenance strategies** (corrective, preventive, predictive, and improvement maintenance).
- Illustrated real-world applications through case studies from **Heineken Bralima Group**, emphasizing production continuity and operational safety.
- Defined the **role and responsibilities of maintenance engineers** in optimizing equipment reliability and reducing downtime.
- Supervised and coordinated maintenance activities to ensure compliance with **safety, quality, and environmental standards**.
- Introduced **CMMS (Computerized Maintenance Management Systems)** for digital planning, tracking, and optimization of maintenance operations.
- Promoted a **data-driven maintenance culture**, integrating condition monitoring, vibration analysis, and performance indicators (MTBF, MTTR).
- **Co-supervised Bachelor's and Master's theses**, providing academic guidance, technical expertise, and methodological support in experimental testing, data-driven analysis, and predictive modeling.
- Facilitated **laboratory activities** while ensuring compliance with academic and safety standards.

 **Institut Supérieur des Techniques Appliquées (ISTA/Kinshasa)** – Kinshasa , Democratic Republic of the Congo

City: Kinshasa | Country: Democratic Republic of the Congo

Lecturer in Production Mechanics and Packaging Systems: Special Course

[01/10/2020 – 01/09/2021]

- Delivered a **specialized course on the beer production process**, focusing on **bottling, packaging machinery, and production line integration** in brewery environments.
- Provided **technical insights into filling systems, labeling units, conveyors, and automation technologies** used in modern bottling plants
- Illustrated **industrial applications through case studies from Heineken Bralima Group**, highlighting efficiency, maintenance, and process optimization.
- Emphasized **industrial safety procedures, risk prevention, and machine compliance** according to international production standards.
- Linked **maintenance strategies, and process control** to enhance students' understanding of **bottling line performance and reliability**.
- Organized and supervised **guided industrial visits** to breweries, allowing students to observe **real production lines**.

EDUCATION AND TRAINING

PhD in Aerospace Engineering and Sciences

Politecnico di Bari [01/11/2021 – 01/10/2025]

City: Bari | Country: Italy | Final grade: Excellent | Level in EQF: EQF level 8 | Thesis: Advanced Acoustic Emission Methods for Damage Mechanisms Monitoring in Aerospace Materials

Link: <https://iris.poliba.it/handle/11589/283440>

Master's degree in mechanical engineering (Production)

Institut Supérieur des Techniques Appliquées (ISTA/Kinshasa) [01/10/2016 – 11/10/2016]

City: Kinshasa | Country: Democratic Republic of the Congo | Final grade: Distinction 70% | Level in EQF: EQF level 7 | Thesis: Analyse vibratoire de dégradations mécaniques: Surveillance de l'usure d'outils de coupe en tournage mécanique par analyse vibratoire

Link: <https://www.amazon.com/dp/6138497988>

Bachelor's degree in Electromechanics

Institut Supérieur des Techniques Appliquées (ISTA/Kinshasa) [01/10/2013 – 30/09/2016]

City: Kinshasa | Country: Democratic Republic of the Congo | Level in EQF: EQF level 6

LANGUAGE SKILLS

Mother tongue(s): French | Swahili | Lingala | Tshiluba

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Italian

LISTENING B1 **READING** A2 **WRITING** B1

SPOKEN PRODUCTION B1 **SPOKEN INTERACTION** B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Acoustic Emission Testing / Python-Programming / Advanced MATLAB user / AI, Deep Learning / Structural Health Monitoring / Vibration Analysis / sensor integration / High Frequency / Data shaping (data acquisition, transformation and cleansing) / Analog and Digital Signals Processing / Uncertainty Quantification / Condition monitoring / Machine Learning • / Polymer Composites / Characterizing the structures of materials

PUBLICATIONS

[2025]

Intelligent monitoring of bolt loosening in industrial customized

Authors: S. Carone, C. Casavola, D. Katamba Mpoyi, and G. Pappalettera | **Publisher:** Measurement Science and Technology

[2005]

Deep Autoencoder Framework for Classifying Damage Mechanisms in Repaired CFRP

Authors: Barile, C.; Casavola, C.; Katamba Mpoyi, D.; Pappalettera, G. | **Publisher:** Appl. Sc

[2023]

Acoustic Emission and Deep Learning for the Classification of the Mechanical Behavior of AlSi10Mg

Publisher: Applied Sciences

[2024]

Wear Monitoring Based on Vibration Measurement During Machining: An Application of FDM and EMD

Authors: Mpoyi Dany Katamba, Ekuakille, Aimé Lay, Ugwiri Moise Avoci, Casavola C and Pappalettera G | **Publisher:** Measurement: Sensors

[2024]

Enhancing the AET Using Fuzzy Artificial Bee Colony Based DL for Characterizing SLM AlSi10Mg

Authors: Barile C, Katamba Mpoyi D, Casavola, C, Pappalettera G., and Paramsamy Kannan Vimalathithan. | **Publisher:** International Journal of Damage Mechanics

[2019]

Application of empirical mode decomposition for tool wear monitoring using vibration measurement

Authors: Mpoyi Katamba D, Lay-Ekuakille A, Kidiamboko Guwa Gwa Band S | **Publisher:** 8th Env Imeko - IMEKO TC19 Symposium

CONFERENCES AND SEMINARS

[04/09/2024 – 06/09/2024] Napoli

53° Convegno Nazionale AIAS *Damage Characterization of Adhesively Bonded CFRP Composites Using Acoustic Emission Signals and Machine Learning*. Oral Presentation.

[15/05/2024 – 17/05/2024] Bari

New Trends in Fatigue and Fracture 2024 *Monitoring the Mechanical Behavior of AlSi10Mg AM-SLM Specimens Using Acoustic Emission and Fuzzy Artificial Bee Colony-based Deep Learning*. Oral Presentation.

[24/08/2022 – 26/08/2022] online

MeSSAC 2022: Measurement, Sensor Systems and Applications Conference Wear

Monitoring Based on Vibration Measurement During Machining: An Application of FDM and EMD. Oral Presentation.

[29/10/2019 – 30/10/2019] Sfax, Tunisia

8th Envimeko - IMEKO TC19 Symposium on Environmental Instrumentation and Measurements *Application of Empirical Mode Decomposition for Tool Wear Monitoring Using Vibration Measurement*. Oral Presentation.

VOLUNTEERING

[09/04/2025 – Current]

Materials Research Express: Reviewer

[11/10/2024 – Current]

MDPI Acoustics: Reviewer

[17/04/2024 – Current]

MDPI Applied Sciences: Reviewer

[22/11/2023 – Current]

Measurement Sensors: Reviewer

[01/06/2023 – Current]

Measurement : Reviewer

[05/10/2022 – Current]

IEEE Sensors Journal : Reviewer